ECONOMETRICS I
Practice Problem Set 2

The textbook is Jeffrey M. Wooldridge Introductory Econometrics: A Modern Approach, 6th ed., 2016, Cengage Learning

PROBLEMS

1. A random sample of working men was used to estimate the following equation:

$$\widehat{educ} = 10.36 - 0.094sibs + 0.131meduc + 0.210feduc$$

$$n = 722, R^2 = 0.214,$$

where *educ* is years of schooling, *sibs* is number of siblings, *meduc* is mother 's years of schooling, and *feduc* is father 's years of schooling.

   (a) Does *sibs* have the expected effect? Explain. Holding *meduc* and *feduc* fixed, by how much does *sibs* have to increase to reduce predicted years of education by one year? (A noninteger answer is acceptable here.)

   (b) Discuss the interpretation of the coefficient on *meduc*

   (c) Suppose that Man A has no siblings, and his mother and father each have 12 years of education. Man B has no siblings, and his mother and father each have 16 years of education. What is the predicted difference in years of education between B and A?

2. In a study relating college grade point average to time spent in various activities, you distribute a survey to several students. The students are asked how many hours they spend each week in four activities: studying, sleeping, working, and leisure. Any activity is put into one of the four categories, so that for each student the sum of hours in the four activities must be 168.

   (a) In the model

$$GPA = \beta_0 + \beta_1 study + \beta_2 sleep + \beta_3 work + \beta_4 leisure + u,$$

   does it make sense to hold *sleep*, *work*, and *leisure* fixed, while changing *study*?

   (b) Explain why this model violates No perfect collinearity assumption.

   (c) How could you reformulate the model so that its parameters have a useful interpretation and it satisfies no perfect multicollinearity assumption?

3. Consider an equation to explain salaries of CEOs in terms of annual firm sales, return on equity (roe, in percentage form), and return on the firm's stock (ros, in percentage form):

$$\log(salary) = \beta_0 + \beta_1 \log(sales) + \beta_2 roe + \beta_3 ros + u.$$

   (a) In terms of the model parameters, state the null hypothesis that, after controlling for *sales* and *roe*, *ros* has no effect on CEO salary. State the alternative that better stock market performance increases a CEO's salary.

   (b) Using the data in CEOSAL1, the following equation was obtained by OLS:

$$\widehat{\log(salary)} = \underset{(.32)}{4.32} + \underset{(.035)}{.280}\log(sales) + \underset{(.0041)}{.0174}roe + \underset{(.00054)}{.00024}ros$$
$$n = 209, R^2 = .283.$$

   By what percentage is *salary* predicted to increase if *ros* increases by 50 points? Does *ros* have a practically large effect on salary ?

(c) Test the null hypothesis that *ros* has no effect on *salary* against the alternative that *ros* has a positive effect. Carry out the test at the 10% significance level.

(d) Would you include *ros* in a final model explaining CEO compensation in terms of firm performance? Explain.

4. Consider the estimated equation from Example 4.3 in textbook, which can be used to study the effects of skipping class on college GPA:

$$\widehat{colGPA} = \underset{(.33)}{1.39} + \underset{(.094)}{.412}\,hsGPA + \underset{(.011)}{.015}\,ACT - \underset{(.026)}{.083}\,skipped$$
$$n = 141, R^2 = .234.$$

(a) Using the standard normal approximation, find the 95% confidence interval for β_{hsGPA}.

(b) Can you reject the hypothesis $H_0 : \beta_{hsGPA} = .4$ against the two-sided alternative at the 5% level?

(c) Can you reject the hypothesis $H_0 : \beta_{hsGPA} = 1$ against the two-sided alternative at the 5% level?

5. In Problem 3 in Chapter 3 of textbook, we estimated the equation

$$\widehat{sleep} = \underset{(112.28)}{3638.25} - \underset{(.017)}{.148}\,totwrk - \underset{(5.88)}{11.13}\,educ + \underset{(1.45)}{2.20}\,age$$
$$n = 706, R^2 = .113.$$

where we now report standard errors along with the estimates.

(a) Is either *educ* or *age* individually significant at the 5% level against a two-sided alternative? Show your work.

(b) Dropping *educ* and *age* from the equation gives

$$\widehat{sleep} = \underset{(38.91)}{3586.38} - \underset{(.017)}{.151}\,totwrk$$
$$n = 706, R^2 = .103.$$

Are *educ* and *age* jointly significant in the original equation at the 5% level? Justify your answer.

(c) Does including *educ* and *age* in the model greatly affect the estimated tradeoff between sleeping and working?

(d) Suppose that the sleep equation contains heteroskedasticity. What does this mean about the tests computed in parts **(a)** and **(b)**?

6. Consider the following R output

```
> res<-lm(log(rd)~ log(sales)+profmarg, data=rdchem)
> summary(res)
Coefficients:
             Estimate  Std.  Error  t value  Pr(>|t|)
 (Intercept) -4.37827    0.46802    -9.355  2.93e-10
 log(sales)   1.08422    0.06020     ......   ......
  profmarg    0.02166    0.01278     1.694    0.101


Residual standard error:  0.5136 on 29 degrees of freedom
Multiple R-squared:  0.918, Adjusted R-squared:  0.9123
F-statistic:  ......  on 2 and 29 DF
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In this regression output, rd is the R&D expenditures of the firms, sales is the total annual sales, and profmarg is the profits as a percentage of sales.

(a) Write the SRF in equation form.

(b) Find the t-statistic on log(sales). Test the statistical significance of log(sales) at $\alpha = 0.01$ (two sided critical value is 2.76). Can you find an upper bound on the p-value? (hint: fill in the following $p - value < \ldots\ldots$).

(c) Is profitmarg statistically significant at 5%? How about 10%? 11%?

(d) Test the overall statistical significane of the regression using the F statistic. (critical value is $c = 3.33$ at 5%level).

Review Questions:
(These will not be solved. Read the classnotes and the textbook)

7. State the Gauss-Markov Theorem and list the required assumptions. Briefly explain.

8. Use your knowledge from Microeconomics classes to specify an econometric model for the Cobb-Douglas production function with output (Q) as the dependent variable, and Capital (K) and Labor (L) as the explanatory variables.What factors are included in the error term? Be specific.

9. Briefly discuss the finite sample properties of OLS estimators when an important variable is omitted.